

# Yu-Yen Chang

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Experienced data science researcher with a PhD in astrophysics and 15+ years of experience in scientific computing and data-driven research. Proficient in statistics, modeling, machine learning, computer vision, and programming, with a proven track record of solving complex problems across scientific and engineering domains.

## Experience

- **Senior Data/Computational Scientist**, Turion Space, USA 01/2025 - Present
  - Lead the development of ML and computer vision solutions for satellite imagery and synthetic data.
  - Develop 10+ advanced algorithms for detection, tracking, registration, and estimation.
  - Design multi-modal data pipelines integrating multi-spectral, temporal, and contextual data.
  - Apply ML, physics-informed ML, and numerical optimization to complex space and engineering problems.
  - Develop scalable data and algorithm pipelines and collaborate across research and software teams.
- **Associate/Assistant Professor**, Physics, NCHU, Taiwan 02/2020 - 01/2025
  - Led research group in classifying, clustering, and predicting diverse data with 80%+ performance.
  - Achieved 85%+ precision using cloud-based computing to investigate physical properties of galaxies.
  - Analyzed, simulated, and visualized 1M+ datasets from 30+ databases to support collaborative research.
  - Mentored 10+ students and postdoctoral researchers in astrophysics and data science.
  - Delivered 10+ lectures and 50+ public presentations, and published 40+ papers with 6,000+ citations.
- **Insight Data Science Fellow** (20A.DS.SV), INSIGHT, USA 01/2020 - 03/2020
  - Discovered societal concerns from unstructured 100K+ survey responses with Elucd.
  - Provided location-specific living information for customers with 80%+ performance.
  - Utilized Natural Language Processing to cluster unstructured comments using text and geographic information.
- **Postdoctoral Research Fellow**, ASIAA, Taiwan 10/2016 - 01/2020
  - Made breakthroughs in understanding active galaxies by XGBoost and Keras algorithms.
  - Interpreted different types of super-massive black holes and reached 80%+ performance.
  - Inspected galaxy evolution using 30+ bands and 20+ datasets across multiple wavelengths.
- **Postdoctoral Research Fellow**, CEA-Saclay, France 10/2014 - 09/2016
  - Created a catalog for 500K+ distant galaxies with state-of-the-art decomposed models.
  - Evaluated sizes differences ( $5\sigma+$ ) between active and non-active galaxies using image processing.
  - Obtained observational data through successful telescope proposals and analyzed galaxy morphology statistically.
- **Postdoctoral/Graduate Research Fellow**, MPIA, Germany 09/2010 - 06/2014
  - Generated an 800K+ galaxy catalog by implementing Bayesian and chi-square fitting techniques.
  - Performed 3D reconstruction of galaxy shapes from 2D projected images.
  - Modeled structural/size evolution, identifying differences in nearby/distant targets ( $P_{KS} < 0.05$ ) over 11 Gyr.

## Education

- **Ph.D.:** Astronomy, Heidelberg University, Germany 12/2013
- **M.S.:** Physics, National Taiwan University, Taiwan 06/2010
- **M.S.:** Radio Astronomy and Space Science, Chalmers University of Technology, Sweden 01/2009
- **B.S.:** Communication Engineering, National Chiao-Tung University, Taiwan 06/2007

## Skills

- Machine Learning, Deep Learning, Computer Vision, Statistical Modeling
- Image Processing, Object Detection, Object Tracking, Image Registration, 3D Reconstruction
- Numerical Optimization, Scientific Computing, Simulation, Bayesian Inference, Time-Series Analysis
- Satellite Imagery, Geospatial Data, Multi-spectral Data, Synthetic Data, Physics-informed ML
- Data Pipelines, Distributed Computing, Data Modeling, Cloud Computing
- MLOps, Model Deployment, Model Serving, Model Monitoring, Docker
- Programming: Python, SQL, C, C++, MATLAB, IDL, Mathematica, Bash, Git, Linux
- ML/Data Tools: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, NumPy, Pandas, SciPy